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Exploring the Impact of Hearing Loss on Depression and Quality of Life

Sara Gibert Medina* and Júlia Escoriza Pons†

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This study examines the association between self-reported hearing condition and depressive symptoms using data from the National Health and Nutrition Examination Survey (NHANES). Depression was measured using two Patient Health Questionnaire items (DPQ010 and DPQ020), and hearing status was the main predictor. Correlation analysis, multiple linear regression models, ANOVA, and t-tests were applied, adjusting for age, household income, and smoking habits. Across all analyses, worse hearing condition was significantly associated with higher depressive symptom scores. These results indicate that hearing difficulties are independently linked to depressive symptoms in the U.S. population.

Keywords: Hearing loss, depression, NHANES, mental health, regression analysis

1 Introduction

Hearing impairment is a common chronic condition that affects communication, social participation, and overall quality of life. Beyond its functional limitations, hearing loss has been increasingly linked to adverse mental health outcomes, particularly depressive symptoms, potentially due to social isolation and reduced interpersonal engagement. However, empirical evidence quantifying this association remains limited and often insufficiently controlled for key demographic and socioeconomic factors.

This study examines the relationship between self-reported hearing condition and depressive symptoms using data from the National Health and Nutrition Examination Survey (NHANES). Depression is measured using standardized questionnaire items, and the analysis applies correlation methods, regression models, and group comparisons while adjusting for age, household income, and smoking habits. The central hypothesis is that poorer hearing condition is significantly associated with higher levels of depressive symptoms in the U.S. population.

2 Data and Methods

2.1 Data Source

Data were obtained from the National Health and Nutrition Examination Survey (NHANES). NHANES combines interviews, physical examinations, and standardized questionnaires to assess the health and nutritional status of the U.S. population.

This study integrates multiple NHANES questionnaire modules using the unique participant identifier (SEQN). The analysis focuses on adult participants with complete information on hearing condition, depressive symptoms, and relevant covariates. NHANES sampling weights and complex survey design were not applied, as the objective of this study was to examine associations rather than produce population-level estimates.

2.2 Variables

The primary independent variable was self-reported hearing condition, derived from NHANES auditory questionnaire item AUQ054. Hearing status was treated as an ordinal variable, with higher values indicating worse hearing ability.

Depressive symptoms were measured using two items from the Patient Health Questionnaire (PHQ-2): DPQ010 (lack of interest or pleasure in doing things) and DPQ020 (feeling down, depressed, or hopeless). Responses were coded on an ordinal scale, with higher values reflecting greater symptom frequency.

Covariates were selected based on prior literature and theoretical relevance. Age (RIDAGEYR) was included as a continuous demographic control. Annual household income (INDHHINC) was used as a socioeconomic indicator. Smoking behavior was represented by age at smoking initiation (SMD030), capturing early exposure to smoking-related health risks.

2.3 Statistical Analysis

Statistical analyses were conducted to evaluate the association between self-reported hearing condition and depressive symptoms. Descriptive visualizations were first generated to explore the distribution of key variables and to assess patterns across hearing condition categories.

Boxplots were constructed to compare depression scores across levels of hearing condition, allowing visualization of central tendency, variability, and potential outliers. Bar plots were used to summarize depressive symptom levels among participants with severe hearing impairment and deafness, facilitating group-based comparisons.

Spearman rank correlation was applied to assess the monotonic association between hearing condition and depression scores. Linear regression models were then estimated with depressive symptoms as the dependent variable. A baseline model included hearing condition as the sole predictor, followed by multivariable models adjusting for age, household income, and smoking behavior.

Group differences in depressive symptoms across hearing condition categories were formally tested using one-way analysis of variance (ANOVA). Post-hoc comparisons were conducted using Tukey's honestly significant

* [redacted]
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difference test. Additionally, independent-samples t-tests were performed after dichotomizing hearing status into good hearing and hearing difficulty. Statistical significance was evaluated at the 0.05 level.

3 Results

3.1 Depressive Symptoms by Hearing Condition

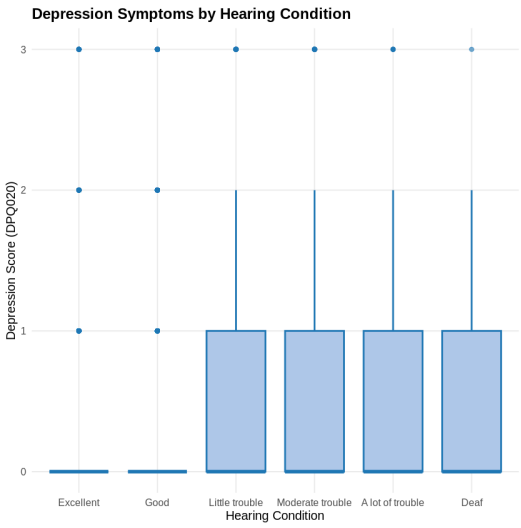


Figure 1. Distribution of DPQ020 depression scores across self-reported hearing condition categories.

Figure 1 presents the distribution of depressive symptom scores (DPQ020) across hearing condition categories. Median depression scores increase progressively as hearing condition worsens. Participants reporting excellent or good hearing exhibit lower median values and narrower interquartile ranges, whereas those reporting hearing difficulties show higher central values and greater variability.

3.2 Group Differences in Depression Scores

Table I. Mean depression scores by self-reported hearing condition

Hearing Condition	Mean DPQ010	Mean DPQ020	N
Excellent	0.27	0.24	1882
Good	0.39	0.34	2003
Little trouble	0.56	0.50	730
Moderate trouble	0.54	0.61	321
A lot of trouble	0.66	0.53	128
Deaf	0.64	0.57	14

Table I summarizes mean depressive symptom scores across self-reported hearing condition categories. Both depression indicators (DPQ010 and DPQ020) show a monotonic increase as hearing condition worsens. Participants reporting excellent or good hearing exhibit the lowest mean depression scores, whereas those reporting hearing difficulties or deafness display higher average values.

3.3 Regression Analysis

3.3.a Hearing and Depression Linear Regression

This analysis examined whether hearing difficulties are associated with depression levels and whether hearing status predicts depressive symptoms. A stepwise approach was adopted, beginning with correlation analysis and progressing to regression modeling, to assess both statistical significance and practical relevance.

Correlation was used to account for the discrete nature of the hearing variable. Hearing difficulties were positively associated with depression scores ($\rho = 0.138$, $p < 2.2 \times 10^{-16}$), indicating that poorer hearing corresponds to higher depressive symptoms. However, the small coefficient suggests a weak relationship, motivating further modeling.

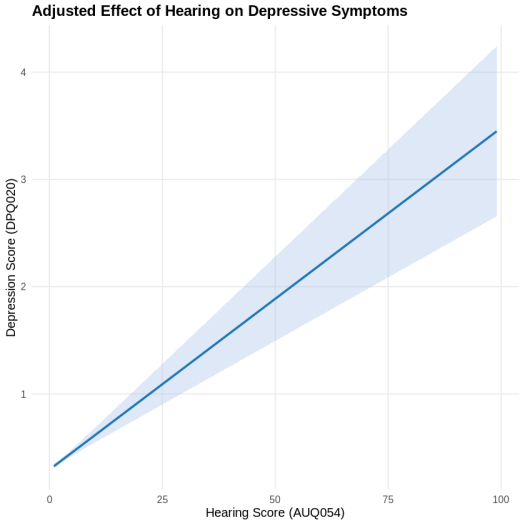


Figure 2. Linear Regression based on hearing condition and Depressive symptoms.

A simple linear regression with depression as the outcome and hearing as the predictor showed a positive and statistically significant association ($\beta = 0.0319$, p ¡ 0.001). Despite significance, explanatory power was limited ($R^2 = 0.011$), indicating that hearing alone explains little variance in depression.

Given the weak explanatory strength and the inherently multivariate structure of NHANES data, additional covariates were considered. Factors such as smoking may confound the relationship between hearing and depression. Therefore, multiple regression models were used to estimate adjusted effects and improve explanatory capacity, representing a necessary progression from bivariate to multivariable analysis.

3.3.b Model 1: Regression based on Age

Model 1 extends the baseline analysis by incorporating age into a multiple linear regression framework to account for demographic confounding. This adjustment is essential given the strong association between hearing difficulties and age, as well as age-related differences in depression.

After controlling for age, hearing status remained a statistically significant predictor of depressive symptoms ($\beta = 0.117$, $p < 0.001$), with a slightly larger effect size than in the simple regression. This finding indicates that the association between hearing difficulties and depression is not explained solely by age and supports the hypothesis that hearing problems are independently associated with higher depression scores.

Age also emerged as a statistically significant predictor, although with a small negative effect (β = -0.0013, $p = 0.033$), suggesting a slight decrease in depressive symptoms with increasing age when hearing status is held constant. While modest in magnitude, this effect justifies age inclusion as a control variable.

Model performance improved relative to the simple regression, with the R^2 increasing from approximately 0.006 to 0.014, more than doubling the explained variance. Although overall explanatory power remained limited, the statistically significant F-statistic confirms that the combined predictors provide a better fit than hearing alone.

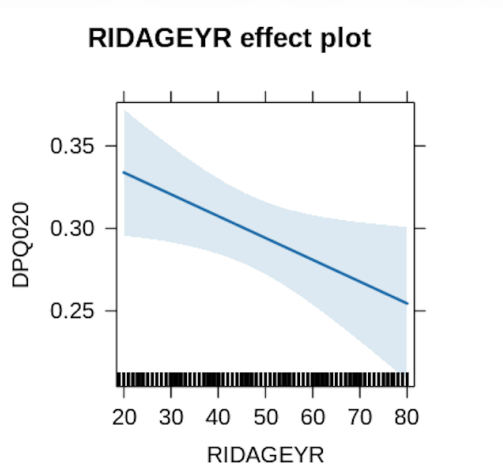


Figure 3. Multiple Regression based on age.

Effect plots corroborate these results. The adjusted hearing effect shows a clear positive association with depression, while the age effect displays a shallow downward trend. Together, these findings demonstrate that hearing difficulties contribute independently to depressive symptoms, with age explaining additional but limited variation.

3.3.c Model 2: Regression based on Household Income

To evaluate whether the hearing–depression association persists across socioeconomic strata, household income was introduced as an additional covariate in Model 2. This approach acknowledges the well-established link between economic resources, health outcomes, and access to care.

The results show that hearing difficulties continue to be positively associated with depressive symptoms after accounting for income ($\beta = 0.028$, $p < 0.001$). Although the magnitude of the hearing coefficient is slightly attenuated relative to the unadjusted model, its continued statistical significance indicates that socioeconomic status does not fully explain the observed relationship.

Household income itself demonstrates a significant inverse association with depression (β =-0.023, $p < 0.001$), with individuals from higher-income households exhibiting lower predicted depression scores. This pattern suggests a protective effect of socioeconomic advantage, independent of hearing status.

Including income substantially improves model fit, with the explained variance increasing to approximately 2.7%. While this value remains modest, it reflects a meaningful gain over simpler specifications and reinforces the multifactorial nature of depressive symptoms. The overall model is statistically significant, confirming the joint contribution of sensory and socioeconomic predictors.

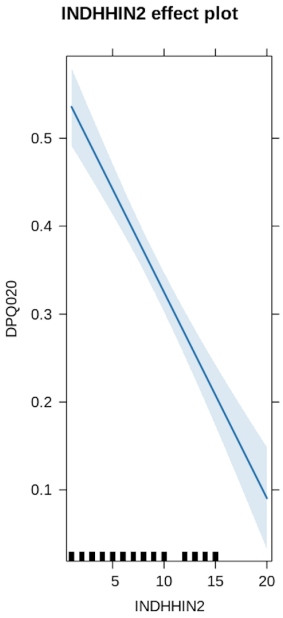


Figure 4. Multiple Regression based on Household Income.

The effect plots provide additional insight into these dynamics. Predicted depression scores rise steadily with increasing hearing difficulties, whereas income is associated with a gradual decline in depressive symptoms across its range. Notably, uncertainty increases at extreme hearing values, reflecting reduced data density, while income effects remain comparatively stable. Together, these visual patterns support the regression results and highlight distinct but complementary roles of hearing and income.

Overall, Model 2 demonstrates that hearing difficulties are linked to higher depressive symptoms independently of household income, while socioeconomic position contributes additional explanatory value. These findings emphasize the importance of considering both sensory health and economic context when modeling depression risk.

3.3.d Model 3: Regression based on Smoking Habits

To assess whether behavioral factors alter the relationship between hearing difficulties and depressive symptoms, Model 3 incorporates smoking exposure as an additional covariate. Smoking was included given its relevance to health behaviors and its potential to co-occur with both sensory impairment and mental health outcomes.

Results indicate that hearing status continues to show a positive and statistically significant association with depressive symptoms after accounting for smoking behavior ($\beta = 0.025$, $p < 0.001$). Although the hearing coefficient is smaller than in the unadjusted model, its persistence suggests that smoking does not fully account for the observed relationship, pointing to an independent contribution of hearing difficulties.

Smoking exposure itself does not exhibit a statistically significant association with depressive symptoms (β =-0.0004, $p = 0.18$). The near-zero estimate and wide uncertainty imply that, within this model specification, smoking adds little explanatory value once hearing status is considered.

From a predictive standpoint, the inclusion of smoking yields only a minor gain in explained variance ($R^2 = 0.008$). This modest improvement highlights the complexity of depressive outcomes and suggests that behavioral variables alone are insufficient to substantially improve model performance.

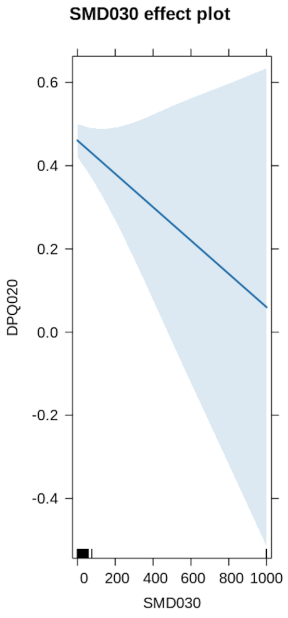


Figure 5. Multiple Regression based on Smoking Habits.

Visual inspection of the effect plots supports these conclusions. Hearing difficulties show a consistent upward trend in predicted depression scores, whereas the smoking effect appears flat and imprecise across its range. Together, these results underscore hearing status as the dominant predictor in Model 3 and indicate that smoking habits play a limited role in explaining depressive symptoms within this dataset.

4 Limitations

Several limitations should be considered when interpreting the findings of this study. First, the analysis is based on cross-sectional data from NHANES, which precludes any causal inference. The observed associations between hearing difficulties and depressive symptoms cannot establish directionality, and reverse causation remains possible.

Second, key variables—including hearing status, depressive symptoms, and smoking habits—were derived from self-reported questionnaires. Self-report measures are subject to recall bias, reporting bias, and misclassification, which may attenuate or inflate observed associations compared to objective assessments.

Third, the explanatory power of the regression models was modest, indicating that a large proportion of variability in depressive symptoms is driven by factors not captured in this analysis, such as social support, comorbid health conditions, cognitive status, or access to healthcare services.

Additionally, some hearing categories, particularly those representing severe hearing impairment, contained relatively small sample sizes. This imbalance may reduce statistical precision and increase uncertainty in effect estimates, especially at extreme values, as reflected in widening confidence intervals.

Finally, although the models adjusted for several important demographic, socioeconomic, and behavioral covariates, residual confounding cannot be ruled out. Unmeasured factors or alternative model specifications may influence the observed relationships.

Despite these limitations, the consistency of results across multiple analytical approaches supports the robustness of the main findings.

5 Conclusion

This study examined the relationship between hearing difficulties and depressive symptoms using NHANES data. Across all analyses, poorer hearing was consistently associated with higher levels of depression, supporting the central hypothesis.

Multivariable models showed that hearing status remained a significant predictor of depressive symptoms after adjustment for age, household income, and smoking behavior. While age and income contributed modestly—older age and higher income being associated with slightly lower depression scores—these factors did not explain the observed hearing–depression relationship. Smoking habits did not show an independent association once hearing status was considered.

Group-based analyses further confirmed that individuals with hearing problems reported significantly higher depressive symptoms than those with good or excellent hearing. Although the models explained a limited proportion of variance, reflecting the multifactorial nature of depression, the findings indicate that hearing difficulties

represent an independent and stable correlate of depressive symptoms.

Overall, these results highlight the importance of considering sensory health in mental health research. Future longitudinal studies using objective hearing measures are needed to better understand causal mechanisms and inform prevention strategies.